

# ■ AgriPredict — Technical Documentation & Architecture Report

**Comprehensive Guide:** How AgriPredict is built, technologies used, feature capabilities, and strategic purpose for Indian Agriculture.

## 1. Executive Overview & Purpose (Why it is Used)

**AgriPredict** is an AI-powered agricultural market intelligence platform designed specifically for Indian farmers, traders, and agronomists. In India, over 140 million farming households face severe price exploitation and information asymmetry due to fragmented local markets, lack of direct APMC mandi price transparency, and aggressive middleman commissions.

### Primary Objectives & Core Purpose:

- **Eliminate Middleman Exploitation:** Provides direct, transparent farm-gate wholesale mandi prices vs. retail market rates across 3,000+ Indian APMCs.
- **Maximize Farmer Profitability:** Identifies the **Highest Profit Mandi (■ Best Choice)** within and near a farmer's district, taking into account arrival volumes and price variances.
- **AI-Driven Decision Making:** Offers 30-day price trend predictions and clear **BUY / SELL / HOLD** recommendations backed by historical data, weather impact, and seasonal demand.
- **Multilingual & Voice Accessibility:** Empowers rural farmers who cannot type in English by supporting voice queries and UI in 6 Indian languages (English, हिन्दी, तेलुगु, मराठी, कन्नड़, ओड़िया).

## 2. Technology Stack & Frameworks (How it is Made)

| Component Layer              | Technology / Service Used              | Function / Purpose                                                                         |
|------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------|
| Frontend Framework           | Next.js 16 (App Router) + React 19     | Server-side rendering, client component state management, fast page prefetching.           |
| Programming Language         | TypeScript (Strict ESNext)             | Type-safe application state, API response validation, zero runtime errors.                 |
| Styling & Responsive Engine  | Tailwind CSS + Custom CSS Variables    | Fluid responsive layout scaling from 320px mobile to 1200px+ desktop frames.               |
| Artificial Intelligence Core | Google Gemini 1.5 Flash API            | Real-time market intelligence, crop trade recommendations, multilingual NLP, document OCR. |
| APMC Market Data API         | Agmarknet Open Data API (data.gov.in)  | Government of India official mandi prices, min/max/modal rates, arrival volumes.           |
| Weather & Soil Radar API     | Open-Meteo Weather API                 | Live temperature, humidity, wind speed, precipitation forecast, soil moisture index.       |
| Geolocation Engine           | OpenStreetMap Nominatim + BigDataCloud | Automatic browser GPS location reverse geocoding to 780+ Indian districts.                 |

|                        |                               |                                                                                          |
|------------------------|-------------------------------|------------------------------------------------------------------------------------------|
| Voice & Speech Engine  | HTML5 Web Speech API          | Speech-to-Text in 6 Indian languages + Roman transliterated scripts (Tanglish/Hinglish). |
| Document Scanner / OCR | Canvas API + Tesseract Parser | Extract text from PM-KISAN notices, Kisan Credit Card loan sanctions, Soil cards.        |
| Data Visualization     | Recharts Library              | Interactive 12-month price trend line charts, volatility indices, trade volume metrics.  |

### 3. Detailed Modules & Feature Breakdown

#### 1. APMC Mandi Price Comparison Engine (`MandiComparison.tsx`)

- Covers **all 28 Indian States, 8 Union Territories, and 780+ Districts**.
- Every district contains **at least 10 authentic APMC Mandis & Market Yards** strictly scoped to that district.
- Displays Wholesale Mandi Rates (■/quintal and ■/kg), Retail Market Prices, Arrival Volumes (Tons/day), Price Variance vs Average, and Market Trend (Rising/Stable/Falling).
- Automatically highlights the ■ **Highest Price Mandi** for maximum profit realization.

#### 2. Crop Varieties & Cultivar Grid (`cropData.ts`)

- Supports rich variety datasets for 28 major Indian crops (e.g. Alphonso, Kesar, Totapuri Mangoes; Grand Naine, Red Banana; Basmati 1121 Rice; Guntur S4 Chilli; Nashik Red Onion; Shimla Royal Apple).
- Displays dedicated Mandi Wholesale badges alongside Retail Market badges.

#### 3. AI Recommendation & 30-Day Market Forecast

- Synthesizes weather signals, historical price volatility, and regional supply arrivals to issue clear **BUY, SELL, or HOLD** directives.
- Displays AI Confidence Score (%) and 30-Day Projected Percentage Trajectory (e.g., ↑ +5.2%).

#### 4. Multilingual AI Voice Assistant (`ChatbotVoice.tsx`)

- Conversational assistant supporting 6 languages (English, Tamil, Telugu, Kannada, Malayalam, Hindi).
- Supports voice microphone input via browser speech recognition and understands Romanized transliteration (e.g. 'thakaali vilai enna' -> Tamil Tomato price query).

#### 5. Government Document OCR Scanner (`DocumentOcr.tsx`)

- Analyzes uploaded images/PDFs of PM-KISAN e-KYC notices, Kisan Credit Card (KCC) loans, PMFBY crop insurance policies, and Soil Health Cards.
- Returns structured plain-language summaries and 3-4 actionable steps in 6 languages.

#### 6. Government Schemes Portal (`/schemes`)

- Searchable directory of major central and state agricultural welfare schemes (PM-KISAN, PMFBY, KCC, PMKSY Drip Irrigation, Agri Infrastructure Fund). Includes eligibility requirements and direct application links.

### 4. Data Flow Architecture & Conclusion

**Architecture Workflow:** When a farmer accesses AgriPredict, the browser detects location via OpenStreetMap/BigDataCloud APIs. The client component sends a lightweight request to `/api/mandi-prices`, which queries Agmarknet open data and Google Gemini AI market intelligence. Results are formatted into wholesale vs. retail rates, filtered by the selected district, and rendered instantly with 0ms perceived latency.

**Conclusion:** AgriPredict combines modern web engineering (Next.js 16, React 19) with artificial intelligence and official government market feeds to transform how Indian farmers track, compare, and sell their agricultural produce.